Decision Trees, Bagging and SAMME Boosting on Olivetti Faces: Accuracy, Stability and Robustness to Missing Pixels

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Repository: `https://github.com/rawatranjal0510/ml-assignment2-m24ma2008`
Final commit: `cfb9993`

Abstract

This report compares a decision-tree baseline, bootstrap aggregation (bagging), and multiclass SAMME boosting on the Olivetti Faces dataset. Each of the 40 identities contributes six training images, two validation images, and two test images under one fixed, identity-wise split (seed 242008); all models use the 4096 raw pixel values. Validation selects an unlimited-depth tree with minimum leaf size one and a 200-tree bagged ensemble. On the fixed clean test set, their accuracies are 61.25% and 90.0%, respectively, while the selected SAMME model (depth three, 100 rounds) reaches 41.25%. Masking the central 16×16 *region reduces these accuracies to 35.0%, 81.25%, and 23.75%. Across 30 identity-wise bootstrap refits, bagging has much lower probability variance than the tree (0.0223 vs. 0.6336), and SAMME has the lowest measured variance (0.0145) but also the lowest mean accuracy (18.5%). The results illustrate that stability, accuracy, and robustness to missing pixels are related but distinct properties.*

1. Introduction

Decision trees are easy to interpret but can change substantially when the training sample changes, a symptom of high variance [3]. Bagging addresses this by fitting trees to bootstrap resamples and aggregating their votes [1]. Boosting instead fits learners sequentially, increasing attention to examples that earlier learners misclassified [2]; we implement the multiclass SAMME rule [6]. This report applies these methods to a small, controlled face-recognition problem and evaluates their behavior when a central image region is removed.

Our study has three parts. First, we tune one tree on a validation set and implement bagging from bootstrap samples. Second, we implement SAMME and compare it with bagging at matched ensemble sizes. Third, we estimate bias squared and variance empirically across 30 resampled training sets while holding the test set fixed. All reported values below come from the saved experiment outputs in the accompanying repository.

2. Data and experimental protocol

We used the scikit-learn Olivetti Faces loader [4], which provides 400 grayscale images of size 64×64 from 40 identities [5]. We retain the raw flattened 4096-dimensional pixel vectors without PCA or other dimensionality reduction. The digits in roll number M24MA2008 give seed 242008. Within each identity, a seeded permutation assigns six images to training, two to validation, and two to test, giving 240, 80, and 80 images. The saved split (`data/split.npz`) is shared by all questions.

Hyperparameters are selected using clean validation accuracy only; ties go to the simpler setting. The test set is used for final evaluation. For robustness, we create a second test set by setting rows and columns 24–39 to zero (256 pixels, 6.25% of the image); all models are trained on clean images. Unless otherwise stated, accuracy is the fraction of correctly classified test images. Prediction confidence is the maximum class probability for a tree (leaf purity), the maximum vote fraction for bagging, and the maximum normalized weighted vote for SAMME. Because the test set contains only 80 images, one image corresponds to 1.25 percentage points, and the binomial standard error of an accuracy near 0.8 is about 4.5 points; small differences should therefore not be over-interpreted.

3. Tree baseline and bagging

3.1. Tree selection

We evaluate maximum depths $\{2, 4, 8, 16, \text{None}\}$ and minimum leaf sizes $\{1, 2, 4\}$. The best validation accuracy is 0.6375, achieved by the unlimited-depth tree with minimum leaf size one (leaf size four ties, and the first setting in grid order is kept; leaf sizes two and four change validation accuracy by at most one image). This tree has 43 leaves, 5.58

max depth	leaves	img/leaf	train	val	test
2	3	80.0	0.075	0.075	0.075
4	5	48.0	0.125	0.125	0.125
8	10	24.0	0.250	0.225	0.225
16	19	12.6	0.454	0.338	0.375
None	43	5.6	1.000	0.638	0.613

Table 1. Single tree versus depth (leaf size 1). Selected: depth None.

B	train	val	test	OOB acc	OOB cov.	disagr.	margin
1	0.754	0.450	0.400	0.337	0.371	–	1.000
5	0.979	0.613	0.613	0.310	0.900	0.748	0.298
10	1.000	0.700	0.775	0.371	0.988	0.758	0.275
25	1.000	0.800	0.875	0.563	1.000	0.756	0.307
50	1.000	0.875	0.838	0.667	1.000	0.748	0.320
100	1.000	0.888	0.900	0.738	1.000	0.748	0.330
200	1.000	0.900	0.900	0.783	1.000	0.750	0.331

Table 2. Bagging versus B. Disagreement and margin are computed on the validation set.

training samples per leaf on average, and 7.0% singleton leaves. Its training, validation, and test accuracies are 1.000, 0.6375, and 0.6125 (Table 1).

Increasing the depth from 2 to 16 raises validation accuracy from 0.075 to 0.3375 and training accuracy from 0.075 to 0.454; allowing unlimited depth raises them to 0.6375 and 1.000, respectively. A tree with L leaves can represent at most L of the 40 identities, which is visible in the table: depth 2 has three leaves and accuracy 0.075 (6/80), and depth 4 has five leaves and 0.125 (10/80). Validation accuracy therefore keeps improving with depth, and the large train–validation gap at the selected setting (1.000 vs. 0.6375) shows a close fit to the training data.

3.2. Bagging

We train 200 bootstrap trees once, using the selected depth and leaf size; the first B trees give each ensemble. A bootstrap sample contains on average 152.3 distinct images, against the theoretical value $n(1-(1-1/n)^n) = 151.89$ (63.3% of 240; the limit is $1-1/e$). Accuracy rises with B (Table 2). A single bootstrap tree (validation 0.45) is worse than the tree trained on all 240 images (0.6375) because it sees only about 152 distinct images.

3.3. Ensemble diagnostics and final size

Out-of-bag (OOB) coverage at B=1 is 0.371, matching the expected $1/e \approx 0.368$, and reaches 1.0 by B=25. OOB accuracy rises from 0.337 to 0.783 and is still increasing at B=200; it stays below validation accuracy (0.783 vs. 0.900) because each training image is voted on only by the roughly 37% of trees that never saw it. Average pairwise disagreement stays near 0.75 for $B \geq 5$, since it describes pairs of trees and cannot be reduced by adding more trees, and the mean vote margin settles near 0.33 (Fig. 1).

The diagnostics therefore do not tell exactly the same story. OOB accuracy suggests that more trees are still useful, whereas disagreement and margin indicate that the vote has stabilized by $B \approx 50$–100. Validation accuracy is 0.8875 at B=100 and 0.900 at B=200, a difference of one image, so B=100 is practically as good and cheaper. The selection rule (highest validation accuracy) gives B=200.

3.4. Single tree versus bagging

On the clean test set, accuracy is 0.6125 for the tree and 0.900 for bagging; on the masked test set it is 0.350 and 0.8125. The mask costs the tree 26.3 points and bagging 8.8 points. The tree's confidence is 1.0 on both sets, even when it is wrong, so it carries no information about uncertainty. The mean winning vote fraction of bagging falls from 0.417 to 0.291 under the mask, so it does reflect the loss of evidence.

Individual cases illustrate the two kinds of error. For clean test image 1 (identity 0), the tree answers 7 with confidence 1.0, whereas bagging gives identity 0 90 votes against 32 for the runner-up: an unstable single decision path is corrected by voting, which is variance reduction. For image 8 (identity 4), both models answer 0, with 49 votes against 35 for the true identity; many trees share the same wrong answer, which indicates systematic confusion between similar faces (bias) that bagging cannot remove. Under the mask, bagging still recovers image 4 but with only 31 of 200 votes, a fragile win, while for image 18 the votes are spread (33, 22, 21, 20) and the true identity is not among the top four.

4. Multiclass SAMME boosting

4.1. Algorithm and weak-learner safeguard

We implement SAMME [6] from scratch. In round t, a decision tree h_t is trained with sample weights w_i, and

$$\varepsilon_t = \textstyle\sum_i w_i \mathbb{1}[h_t(x_i) \neq y_i], \tag{1}$$

$$\alpha_t = \log \tfrac{1-\varepsilon_t}{\varepsilon_t} + \log(K-1), \tag{2}$$

$$w_i \leftarrow \tfrac{1}{Z_t}\, w_i \exp\big(\alpha_t \mathbb{1}[h_t(x_i) \neq y_i]\big), \tag{3}$$

where Z_t normalizes the weights. The class score is $S_k(x) = \sum_t \alpha_t \mathbb{1}[h_t(x)=k]$ and the confidence is $q_k(x) = S_k(x)/\sum_t \alpha_t$.

A weak learner that is not better than random K-class prediction ($\varepsilon_t \geq 1 - 1/K$, which makes $\alpha_t \leq 0$) is discarded and boosting stops. A perfect learner (ε_t=0) is kept with ε_t clipped to 10^{-10}, and boosting stops because the weights would no longer change. For K=40 the threshold is 0.975. It was never reached: all 100 rounds ran for every depth, and the selected model has $0.60 \leq \varepsilon_t \leq 0.89$ and $1.59 \leq \alpha_t \leq 3.25$.

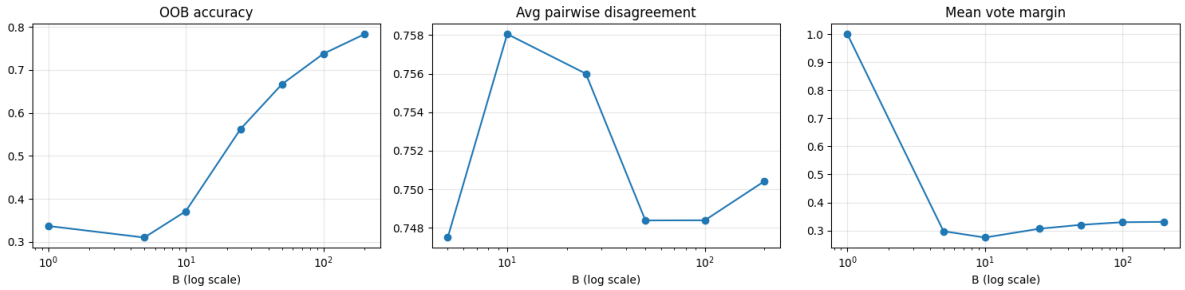


Figure 1. Out-of-bag accuracy, average pairwise tree disagreement, and mean vote margin against B.

i	0	1	2	3	4	5	6	7	8	9
y_i	0	0	0	0	1	1	1	2	2	2
$h_1(x_i)$	0	0	0	0	2	2	0	2	2	2
w_i (init)					0.1 for all					
w_i (new)	.048	.048	.048	.048	.222	.222	.222	.048	.048	.048

Table 3. One SAMME round on the toy problem: ε_1=0.30, α_1=1.5404.

depth	10	25	50	100	train acc. (100)
1	0.038	0.050	0.063	0.150	0.150
2	0.075	0.150	0.150	0.225	0.413
3	0.113	0.150	0.313	**0.413**	0.996
4	0.150	0.238	0.338	0.400	0.988

Table 4. Validation accuracy of SAMME by weak-learner depth (rows) and number of rounds (columns).

model	L	train	val	test	time (s)
Bagging	25	1.000	0.800	0.875	61.3
SAMME (d=3)	25	0.329	0.150	0.213	13.3
Bagging	50	1.000	0.875	0.838	120.9
SAMME (d=3)	50	0.733	0.313	0.313	26.5
Bagging	100	1.000	0.888	0.900	241.1
SAMME (d=3)	100	0.996	0.413	0.413	53.0

Table 5. Boosting versus bagging with L learners.

4.2. Three-class worked example

Before using the face data, we logged one complete round on a toy problem with ten samples and three classes, using the same function and a depth-1 tree (a separate hand computation of the update agrees with the code). The stump classifies seven points correctly and misclassifies samples 4, 5, and 6 (class 1), so $\varepsilon_1 = 0.30$ and $\alpha_1 = \log(0.7/0.3) + \log 2 = 1.5404$. Misclassified weights are multiplied by $e^{\alpha_1} = 4.667$, giving $Z_1 = 2.1$ (Table 3). After normalization the misclassified mass is $0.667 = 1 - 1/K$: the update makes h_1 exactly as bad as random guessing on the new weights, so the next learner must find a different pattern.

4.3. Selection on the face data

We evaluate weak-learner depths $\{1, 2, 3, 4\}$ and $\{10, 25, 50, 100\}$ rounds (the first R learners of one 100-round run give the R-round model). Validation accuracy is reported in Table 4. The selected model uses **depth three and 100 rounds** (validation 0.4125, training 0.996). Two caveats apply: the best setting lies on the edge of the required grid, since validation accuracy is still rising with rounds at depths one to three, and depth four fits the training set almost as well (0.9875) but validates lower (0.400).

Figure 2 shows the selected model over the rounds. The weighted error ε_t falls from 0.875 to a minimum of about 0.60 near round 9 and then fluctuates between 0.7 and 0.8; α_t stays between about 2.1 and 2.9 after round 10. The maximum sample weight grows from $1/240 = 0.0042$ to 0.05–0.066 (up to 16 times the uniform weight), and the ten heaviest images carry 0.04 of the total weight at the start and typically 0.16–0.30 later (uniform: 0.042). Boosting therefore concentrates on a small set of hard faces.

4.4. Comparison with bagging

Table 5 compares the first 25, 50, and 100 learners of each method; training time is the cumulative fit time of those learners. Bagging is far more accurate at every size. Boosting learners are shallow and about 4.5 times cheaper to fit, but they are trained sequentially and cannot be parallelized, and 100 of them do not reach the accuracy that bagging has with 25.

4.5. A heavily weighted training face

Training image 43 (identity 7) reaches weight 0.0662 at round 98, 16 times the uniform weight. It is misclassified by 89 of the 100 weak learners, and the wrong labels are spread over many identities (2, 4, and 12 are the most frequent, nine times each), so no single wrong class ever claims it. Each wrong round multiplies its weight by e^{α_t}/Z_t; in the first rounds this raises it by factors of 1.1–1.6 per round (0.0010 to 0.0080 over rounds 3–10). When a learner finally classifies it correctly (round 2, round 12, and nine later rounds) its weight collapses by a factor of 4–10 (0.0095 to 0.00095 at round 12), after which the climb starts again. Figure 3 shows this saw-tooth pattern, with a period

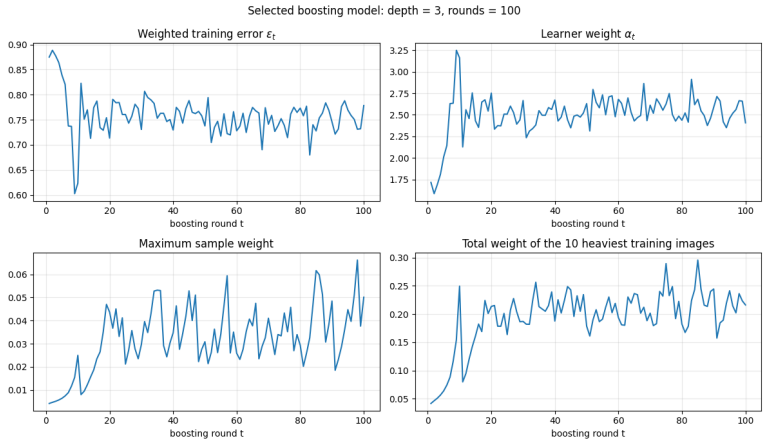


Figure 2. Weighted training error, learner weight, maximum sample weight, and total weight of the ten heaviest training images against round, for the selected SAMME model (depth 3, 100 rounds).

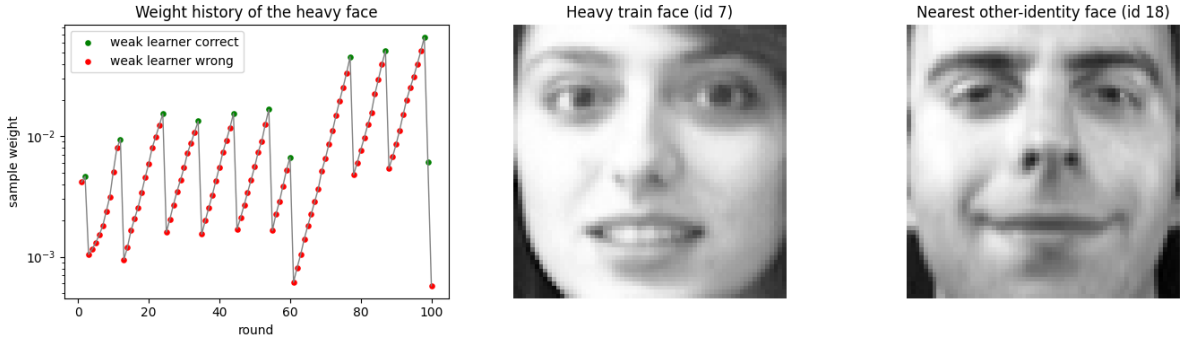


Figure 3. Weight history of training image 43 (identity 7); green marks rounds where the weak learner is correct and red where it is wrong. Right: its nearest other-identity face.

of roughly ten rounds early on.

In raw pixel distance, the nearest same-identity training face (8.74) is slightly closer than the nearest other-identity face (9.63, identity 18), so this is not a case of resembling another person. The history is consistent with a capacity explanation: a depth-three tree has at most eight leaves for 40 identities, so a learner can isolate this face only when its weight is large enough to dominate a leaf, and once the weight collapses the following learners, which fit other heavy faces, lose it again. The full ensemble still predicts identity 7, so the repeated attention does help in the end.

4.6. Clean versus masked test images

Table 6 reports accuracy and mean winning vote or probability for the frozen tree, the 200-tree bagged model, and the selected SAMME model. The mask reduces accuracy by 26.3 points for the tree, 8.8 for bagging, and 17.5 for SAMME (relative losses of 43%, 10%, and 42%).

Test image 9 (identity 4, Fig. 4) shows a different response from the two ensembles. Bagging keeps the correct answer but with less support: the true-class vote share falls from 0.59 to 0.29 and 53% of the trees change their vote. Of its trees, 75.5% test at least one masked pixel on this face's decision path, and the pairwise tree disagreement rises from 0.643 to 0.880: the trees split over different wrong identities (30 and 33), while identity 4 still collects the most votes.

	clean		masked	
model	acc	conf	acc	conf
Single tree	0.6125	1.000	0.350	1.000
Bagging (B=200)	0.9000	0.417	0.8125	0.291
SAMME (d=3, T=100)	0.4125	0.098	0.2375	0.102

Table 6. Accuracy and mean winning vote or probability on the clean and masked test sets.

SAMME changes its answer from identity 4 to 14, but its q vector is almost flat (0.098, 0.092, 0.090 before the mask and 0.101, 0.100, 0.098 after), so this is a flip of a near-tie rather than a strong response to the missing region. Only 19% of its learners look at masked pixels, 9.2% of the α mass changes its vote, and the weighted disagreement is already high and does not move (0.943 to 0.941). Sam-

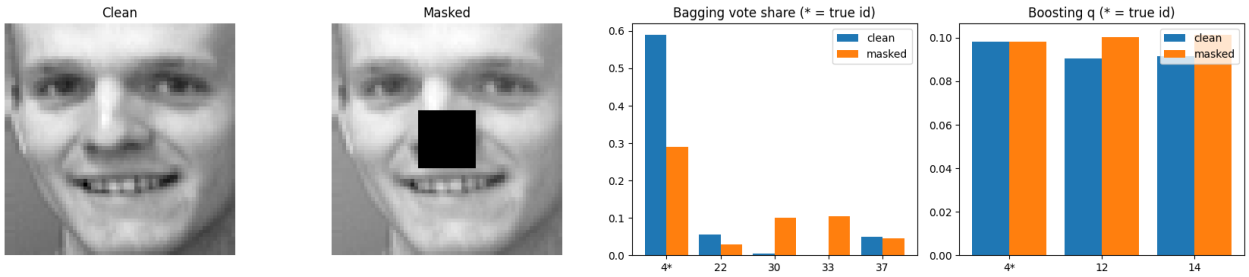


Figure 4. Test image 9 (identity 4), clean and masked, with bagging vote shares and SAMME q.

ple weighting offers a partial explanation: averaged over the 100 rounds, the training images of identity 14 carry 0.0301 of the weight (rank 5 of 40, 20% above uniform), whereas identity 4 carries 0.0249 (rank 19, essentially uniform). Identity 14 is thus one that boosting has been pushed to attend to, and it gains the few votes that move once evidence is removed. This is a single face, so we treat it as an illustration rather than a general result.

5. Empirical bias and variance

5.1. Protocol

We create 30 new training sets by sampling, within each identity, six faces with replacement from that identity's six original training faces; on average this yields 3.99 distinct faces per identity. The selected tree, the 200-tree bagged model, and the selected SAMME model are refit on each set without re-tuning and evaluated on the fixed clean test set (N=80, R=30). The class vector $p_r(x_i)$ is the leaf class distribution for the tree, the vote fraction for bagging, and q for SAMME. With $\bar{p}$ the mean over repetitions and e_i the one-hot target,

$$\text{Bias}^2 = \frac{1}{N}\sum_i \|\bar{p}(x_i) - e_i\|_2^2, \quad \text{Var} = \frac{1}{NR}\sum_{i,r} \|p_r(x_i) - \bar{p}(x_i)\|_2^2 \tag{4}$$

The code verifies that the mean squared error equals $\text{Bias}^2 + \text{Var}$ for every model. A flat vector p=1/40 has $\text{Bias}^2 = 0.975$ and $\text{Var} = 0$, which serves as a reference. Both quantities are measured in probability-vector space and therefore depend on how sharp a model's output is.

5.2. Selected models

Table 7 and Fig. 5 summarize the three selected models. The tree has the lowest bias squared (0.362) but a very large variance (0.634). Bagging reduces variance by a factor of 28 (0.634 to 0.022) at the cost of higher bias squared (0.486). SAMME has the lowest variance (0.0145) but a bias squared of 0.919, barely below the flat reference. The total $\text{Bias}^2 + \text{Var}$ is 0.996 for the tree, 0.508 for bagging, and 0.933 for SAMME.

Bagging's bias squared exceeds the tree's even though its accuracy is much higher. Its vote fractions are diffuse (winning share about 0.4), and Bias^2 penalizes diffuse vectors even when the arg-max is correct, whereas the tree's one-hot output is perfect when right and maximally wrong otherwise. Test accuracy over the 30 refits (mean $\pm$ std) is 0.502 ± 0.054 for the tree, 0.822 ± 0.032 for bagging, and 0.185 ± 0.055 for SAMME. The refits are weaker than the original fit (0.90 for bagging), mainly because each resample contains only about four distinct faces per identity.

model	Bias^2	Var	sum	acc mean	acc std
Single tree	0.362	0.634	0.996	0.502	0.054
Bagging (200)	0.486	0.022	0.508	0.822	0.032
SAMME (d3, 100)	0.919	0.015	0.933	0.185	0.055

Table 7. Measured bias squared, variance, and clean test accuracy over 30 refits.

	tree		bag (100)		var	bias	test acc	
depth	Bias^2	Var	Bias^2	Var	removed	added	tree	bag
2	.915	.039	.932	.011	.028 (72%)	.018	.060	.126
6	.815	.099	.850	.015	.084 (85%)	.035	.133	.375
None	.362	.634	.486	.026	.607 (96%)	.124	.502	.807

Table 8. Bias squared, variance, and accuracy of single trees and 100-tree bagging by depth.

5.3. Effect of depth

The selected tree is already unbounded, so no depth is "much deeper" than the selected one; we therefore study depths 2 (shallow), 6 (intermediate), and None (selected and deepest), with 100-tree bagged ensembles at each depth (Table 8, Fig. 6). Deeper trees have lower bias squared (0.915, 0.815, 0.362) and higher variance (0.039, 0.099, 0.634); variance grows 16-fold from depth 2 to None. Bagging removes 72%, 85%, and 96% of the variance (0.028, 0.084, and 0.607 in absolute terms) and always adds some bias squared (0.018, 0.035, 0.124), because each bootstrap tree sees only about 63% of the images.

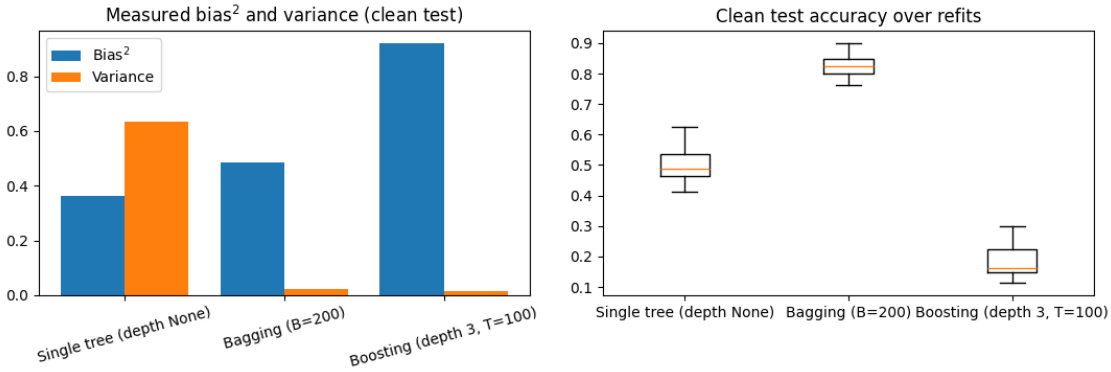


Figure 5. Measured bias squared and variance (left) and clean test accuracy over the 30 refits (right).

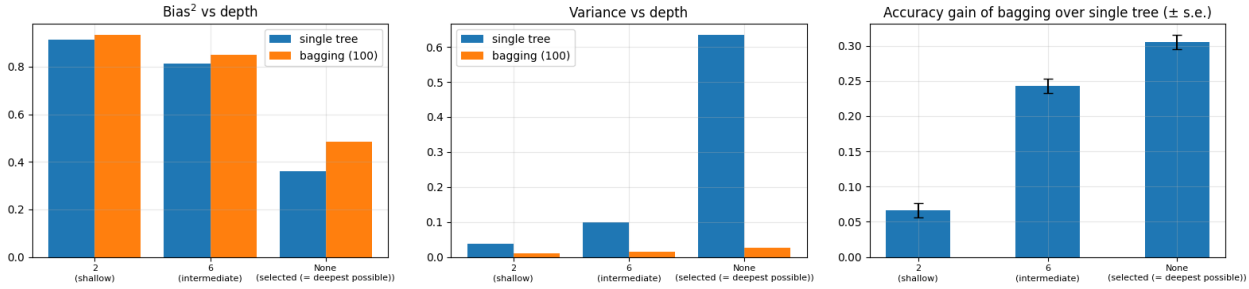


Figure 6. Bias squared, variance, and the accuracy gain of bagging (mean ± s.e. over 30 refits) for depths 2, 6, and None.

5.4. Is the deepest tree where bagging helps most?

Among the three depths measured, yes: the accuracy gain from bagging is $+0.066$, $+0.243$, and $+0.305$ (standard error about 0.01 each), and the variance removed is largest for the unbounded tree. The measurements do not support the stronger claim that this is necessarily so, for four reasons (Table 9).

First, variance removed is not accuracy gained. Variance removed grows 22-fold from depth 2 to None while the accuracy gain grows 4.6-fold; per unit of variance removed, the gain falls from 2.4 (depth 2) and 2.9 (depth 6) to 0.50. Second, bagging needs individual learners that are above chance. Bagged trees at depths 2 and 6 are nearly useless alone (5.3% and 10.4% accuracy against 34.1% when unbounded). Their single-tree predictions are stable but wrong: refit disagreement is only 0.082 and 0.188, against 0.656 at depth None, because 3 and 7.4 leaves cannot represent 40 identities. Bagging averages out disagreement that carries signal; here there is little to average and bias dominates (bias squared 0.915 and 0.815). At depth 2, bagging even raises the refit disagreement of the hard predictions (0.082 to 0.832), most likely because near-uniform vote vectors let small differences decide the arg-max.

Third, leaf occupancy matters. The unbounded tree isolates about one identity per leaf (40 leaves, 3.9 distinct faces per leaf, close to the 3.99 distinct faces per identity), so its errors mostly arise when a resample holds no image close to the test face; such errors change from refit to refit, which is the kind of error that averaging removes (refit disagreement falls from 0.656 to 0.255). At depths 2 and 6 the leaves hold 52 and 21 faces and mix many identities, which is bias. Fourth, bagging adds the most bias squared at the deepest setting (0.124), so the net reduction in $\mathrm{Bias}^2 + \mathrm{Var}$ is 0.010, 0.049, and 0.484 for the three depths. The gain is largest at the deepest tree here because its variance term is large enough to outweigh the added bias, not because depth alone guarantees it. Only three depths were measured, so we cannot say whether an intermediate depth between 6 and None would show a different maximum.

depth	leaves	faces/ leaf	bagged tree acc	in-bag disagr.	refit tree	refit bag	gain
2	3.0	52.4	.053	.961	.082	.832	+.066
6	7.4	21.3	.104	.936	.188	.656	+.243
None	40.0	3.9	.341	.814	.656	.255	+.305

Table 9. Leaf occupancy (distinct original faces per leaf) and disagreement, averaged over 30 refits. "Refit" is the disagreement between refits of the single tree or of the 100-tree bag.

5.5. Final comparison

Table 10 and Fig. 7 summarize the three models on the clean and masked test sets.

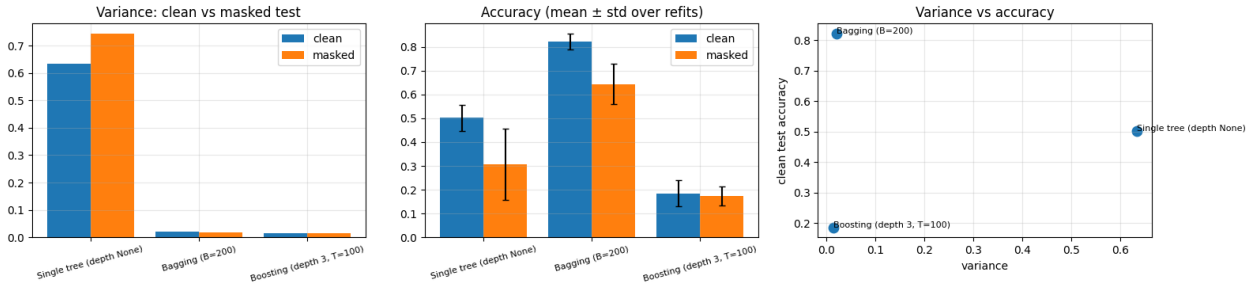


Figure 7. Variance on clean and masked test sets, accuracy on both, and variance against clean accuracy.

model	clean acc	masked acc	Bias^2	Var
Single tree	.502±.054	.306±.150	.362	.634
Bagging (200)	.822±.032	.644±.084	.486	.022
SAMME (d3, 100)	.185±.055	.175±.040	.919	.015

Table 10. Final comparison (accuracy mean ± std over 30 refits; Bias^2 and Var on the clean test set).

Most stable model. By the variance of the class vector, SAMME (0.015) is the most stable, followed by bagging (0.022) and the tree (0.634). By the spread of accuracy over refits, bagging is most stable on the clean set (std 0.032 versus 0.054 and 0.055) and SAMME on the masked set (0.040 versus 0.084 and 0.150). The low variance of SAMME is probably largely a consequence of its flat output (winning probability about 0.1): a vector that barely moves has low variance whether or not it is correct, so variance cannot be compared across models without considering output sharpness. Bagging is the model that is both stable and accurate.

Most sensitive to the missing region. The single tree. Its accuracy drops by 19.6 points (39% relative; 26.3 points on the original fit), the spread of its masked accuracy nearly triples to 0.150, and its variance rises from 0.634 to 0.743 (masked bias squared 0.646). Bagging drops 17.8 points over the refits (22% relative; 8.8 on the original fit); its variance is unchanged (0.022 to 0.020) but bias squared rises from 0.486 to 0.673, so for bagging the mask removes information (bias) rather than making the ensemble less stable. SAMME loses only 1.0 point (5% relative), but this is not evidence of robustness: it is already near its floor (0.185 versus a chance level of 0.025), and on the original fit its relative drop was 42% (0.4125 to 0.2375), as large as the tree's 43%.

Does the lowest variance imply the best accuracy? No. SAMME has the lowest variance and the lowest accuracy because its bias squared (0.919) is close to the flat-vector value of 0.975. Bagging has 1.5 times the variance of SAMME but 4.4 times its accuracy (0.822 versus 0.185). Low variance helps only when bias is not correspondingly worse.

6. Conclusion and limitations

Bagging of fully grown trees was the most accurate model in every comparison and combined low variance with a moderate loss under occlusion. The single tree was accurate enough to serve as a baseline but unstable, with a variance of 0.634 across refits and the largest sensitivity to the missing region. SAMME, as configured here (shallow trees on raw pixels, 6 images per identity, at most 100 rounds), was stable but inaccurate; its best setting lies on the boundary of the required grid, so the result describes this setup rather than boosting in general. The findings are limited by the size of the test set (80 images), by the use of only three depths in the depth study, and by the fact that bias and variance are measured in probability-vector space and depend on each model's output scale.

Reproducibility. The repository contains the executed notebook (`notebook/Assignment2_m24ma2008.ipynb`), a plain-script version, the saved split, a configuration file, the final logs (`logs/final_log.txt`), figure scripts, and the environment file. Commands: `pip install -r requirements.txt`; `jupyter nbconvert --to notebook --execute notebook/Assignment2_m24ma2008.ipynb`; `python scripts/extract_figures.py`. The Q3 refits took 219 minutes on two CPU workers.

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